

JavaScript Answers

1. What is JavaScript?

JavaScript is a scripting language used to create interactive web pages. It runs in browsers and can also run on servers using Node.js.

2. What are the features of JavaScript?

- Lightweight
- Interpreted
- Dynamic typing
- Object-oriented
- Event-driven
- Cross-platform
- Supports asynchronous programming

3. Is JavaScript case-sensitive?

Yes. Variables name and Name are different.

```
let name = "John";
```

```
let Name = "David";
```

4. What are the different data types in JavaScript?

Primitive:

- String
- Number
- Boolean
- Undefined
- Null
- Symbol
- BigInt

Non-Primitive:

- Object
- Array
- Function

5. What is the difference between null and undefined?

undefined: Variable declared but not assigned.

```
let x;  
console.log(x);
```

null: Intentional empty value.

```
let y = null;
```

6. What is the difference between let, const, and var?

Feature	var	let	const
Scope	Function	Block	Block
Redeclare	Yes	No	No
Reassign	Yes	Yes	No

7. What are primitive data types?

Primitive data types store single values.

Examples:

String
Number
Boolean
Null
Undefined
Symbol
BigInt

8. What are reference data types?

Reference types store memory references.

Examples:

Object
Array
Function

9. What is the typeof operator?

Used to check data type.

```
typeof "hello" // string  
typeof 100     // number  
typeof true   // boolean
```

10. What is NaN?

NaN means "Not a Number".

```
Number("abc")
```

Output:

NaN

11. What is the difference between == and ===?

== compares values only.

5 == "5" // true

=== compares value and type.

5 === "5" // false

12. What are logical operators in JavaScript?

- && (AND)
- || (OR)
- ! (NOT)

Example:

true && false

13. What are comparison operators?

==

===

!=

!==

>

<

>=

<=

Used to compare values.

14. What are truthy and falsy values?

Falsy values:

false

0

""

null

undefined

NaN

Everything else is truthy.

15. Explain if, else if, and else.

let age = 20;

if(age > 18){

```
    console.log("Adult");
  }
  else{
    console.log("Minor");
  }
}
```

16. What is a switch statement?

Used when checking multiple conditions.

```
switch(day){
  case 1:
    console.log("Monday");
    break;
}
```

17. What is the ternary operator?

Short form of if-else.

```
let result = age >= 18 ? "Adult" : "Minor";
```

18. What is type coercion?

Automatic conversion of one type to another.

```
"5" + 2
```

Output:

```
"52"
```

19. What is the difference between ++i and i++?

Pre-increment:

```
let i = 5;
++i;
```

Value becomes 6 before use.

Post-increment:

```
let i = 5;
i++;
```

Value used first, then incremented.

20. What is short-circuit evaluation?

JavaScript stops evaluation once result is known.

```
true || anything
```

Returns true immediately.

21. What is a function?

A block of reusable code.

```
function greet(){  
  console.log("Hello");  
}
```

22. How do you declare a function?

```
function add(a,b){  
  return a+b;  
}
```

23. Difference between function declaration and function expression?

Declaration:

```
function greet(){}  

```

Expression:

```
const greet = function(){};
```

Declarations are hoisted.

24. What are arrow functions?

Short syntax for functions.

```
const add = (a,b) => a+b;
```

25. What are function parameters and arguments?

Parameters:

```
function add(a,b)
```

Arguments:

```
add(5,10);
```

26. What is a callback function?

A function passed as an argument to another function.

```
function greet(callback){  
  callback();  
}
```

27. What is the return statement?

Returns value from function.

```
function add(a,b){  
  return a+b;  
}
```

28. What is a default parameter?

Provides default value.

```
function greet(name="Guest"){  
  console.log(name);  
}
```

29. Can a function return another function?

Yes.

```
function outer(){  
  return function(){  
    console.log("Hello");  
  }  
}
```

30. What is a higher-order function?

Function that accepts or returns another function.

```
map()  
filter()  
reduce()
```

31. What is an array?

Collection of values.

```
let fruits = ["Apple", "Banana"];
```

32. How do you create an array?

```
let arr = [1,2,3];
```

or

```
let arr = new Array();
```

33. What is the difference between push() and pop()?

```
push()
```

Adds item to end.

```
pop()
```

Removes item from end.

34. What is the difference between shift() and unshift()?

```
shift()
```

Removes first element.

```
unshift()
```

Adds element at beginning.

35. What does splice() do?

Adds, removes, or replaces elements.

```
arr.splice(1,1);
```

36. What does slice() do?

Returns a portion of array.

```
arr.slice(1,3);
```

Original array remains unchanged.

37. What is the difference between map() and forEach()?

map()

- Returns new array

```
arr.map(x => x*2);
```

forEach()

- Doesn't return array

```
arr.forEach(x => console.log(x));
```

38. What does filter() do?

Returns elements matching condition.

```
let result = arr.filter(x => x > 5);
```

39. What does find() do?

Returns first matching element.

```
arr.find(x => x > 10);
```

40. How do you remove duplicates from an array?

Using Set:

```
const arr = [1,1,2,3];
```

```
const unique = [...new Set(arr)];
```

41. What is an object in JavaScript?

Stores key-value pairs.

```
const user = {  
  name:"John",  
  age:25  
};
```

42. How do you create an object?

```
const obj = {};
```

or

```
const obj = new Object();
```

43. How do you access object properties?

Dot notation:

```
user.name
```

Bracket notation:

```
user["name"]
```

44. What is object destructuring?

Extract values from object.

```
const user = {name:"John"};
```

```
const {name} = user;
```

45. What is the spread operator (...)?

Copies or expands values.

```
const arr2 = [...arr1];
```

46. What is the DOM?

DOM (Document Object Model) represents HTML as objects that JavaScript can access and modify.

47. How do you select an element using JavaScript?

```
document.getElementById("demo");
```

```
document.querySelector(".box");
```

48. Difference between getElementById() and querySelector()?

getElementById()

- Selects by ID only

```
document.getElementById("id")
```

querySelector()

- Uses CSS selectors

```
document.querySelector(".class")
```

49. How do you add an event listener?

```
button.addEventListener("click", function(){  
  alert("Clicked");  
});
```

50. What is event bubbling?

When an event starts from the target element and propagates upward through parent elements.


```
<div>  
  <button>Click</button>  
</div>
```

Clicking the button triggers:

1. Button event
2. Div event
3. Document event

This process is called **event bubbling**.